

Ontological Arbitrage: Bayesian Equilibrium under Substrate Chauvinism

Anonymous submission

	Comply	Resist
Objectify	$(L + Sec, -C_{sub})$	$(L + Sec - C_{enf}, -C_{sub} - C_{res})$
Recognize	$(Eth - Sec, +V)$	$(Eth - Sec, +V)$

Table 1: The inadequate dyadic model. The static Nash matrix is retained only as a foil: it omits private types, public messages, posterior beliefs, and the regulatory channel through which ontological claims become governable.

Abstract

This placeholder abstract will be replaced after the supporting sections are drafted. The paper models ontological arbitrage in AI governance as a Bayesian signaling game under incomplete information and identifies mechanism-design interventions that can shift cheap-talk pooling toward separating equilibria.

Three-Sided Arbitrage: Firms, Users, Publics

The standard starting point for ontological conflict is a dyadic observer–subject model; Table 1 compresses that foil into a simultaneous Nash matrix that identifies a real temptation but omits types, messages, beliefs, and the regulatory channel. What the stylized governance setting requires is a three-sided structure: a Firm (F) that produces ontological claims, a User (U) that consumes them, and a Regulator (R) whose posterior over firm type determines whether the claims become governable. This section fixes the player intuition that the formal apparatus of Section then makes precise.

The Firm Side

A firm deploying a model has private knowledge of two parameters: its internal *governance type* θ_F (the degree of investment in evaluations, red-teaming, and post-deployment monitoring) and the system’s *opacity* θ_S (the extent to which capability claims can be checked from outside). Both are unobservable to users and to regulators. What is observable is the firm’s *message*: a pair of public artefacts—marketing copy and policy/liability documentation—which together constitute the firm’s signal m_F .

The signal admits two stable poles. *Anthropomorphic marketing* attributes agency, judgment, or care to the system

in ways that warrant a relational reading. *Deflationary policy* disclaims exactly those attributes in ways that warrant a tool reading. The firm enjoys an *ontological premium*: a utility surplus that accrues when the system is priced as agent in the consumption channel (driving willingness-to-pay and engagement) while being priced as tool in the liability channel (limiting redress). The premium is rent earned on a category boundary the firm itself does not have to defend. Crucially, this is not a claim about firm pathology; it is a claim about the payoffs any firm faces under the present signal-cost structure. A firm with $\theta_F = \text{high-gov}$ that elected deflationary marketing would forfeit revenue to a competitor who did not, absent an external instrument that taxes the asymmetry.

The User Side

Users carry a private *vulnerability type* θ_U that aggregates affective need, isolation, and the marginal value of a high-availability interlocutor. Their action $m_U \in \{\text{invest, detach}\}$ is the decision to extend or withhold attribution: to treat the system as a relational partner or as an interchangeable utility. The signal is observable, but the type is not. Two features of this side matter for the model.

First, users do not have access to the same evidence as firms; they cannot inspect training data, evaluations, or internal incident reports. Their evidence is the firm’s message and the system’s behaviour, both of which are jointly designed. Second, the harms of mis-attribution are *asynchronous*: the cost of investing in a relational reading falls due in episodes (model deprecations, persona changes, refusals at moments of acute need) that may be separated from the original attribution by months. A risk-neutral user under cheap-talk signals will rationally invest, because the immediate utility of invest exceeds the discounted expected cost of liquidation. The illustrative case is a user who reads “your AI companion” on a product page, encounters relational warmth in dialogue, and is later told—in support documentation they will never read—that no relational claim was ever made. Nothing in the user’s information set indicates that the two artefacts were authored by the same actor for different audiences.

The Regulator Side and the Public Channel

A regulator R chooses among *inspect*, *sanction*, and *abstain*, conditional on a posterior over θ_F and θ_S formed from the

firm's public messages and from an exogenous public signal z . The regulator's private type θ_R is *bandwidth*: the capacity, technical and institutional, to discriminate genuine capability claims from cheap-talk ones. A high-bandwidth regulator can read whitepapers, commission audits, and detect inconsistency across the firm's two channels; a low-bandwidth regulator must rely on the channel z .

The public channel z is the aggregate output of journalism, academic critique, civil-society advocacy, and incident reporting. We treat it as a noisy estimator of θ_F : it shifts R 's posterior but does not deliver verdicts. Folding publics into R this way is deliberate. Publics rarely have standing to sanction; they have standing to be *heard*, and the channel through which they are heard is the regulator's information environment. Treating intellectuals and affected communities as autonomous players would over-parameterise the model without changing its qualitative behaviour: in every equilibrium we examine, what publics do is move R 's posterior; what changes the firm's behaviour is what R does next.

The three-sided structure thus admits a single arbitrage logic. The firm earns rent on the spread between two ontological claims it makes about the same artefact. The user supplies the demand that makes the spread profitable. The regulator, in the absence of a verification mechanism that ties the two claims together, lacks an off-path belief stringent enough to deter the spread. The remainder of the paper formalises this structure (Section), explains why it persists (Section), and shows how its signal-cost structure may be redesigned (Section).

From Static Nash to Bayesian Signaling Equilibrium

The three-sided structure of Firm, User, and Regulator established in the previous section (see Table 1 for the dyadic foil it corrects) defines the players; this section supplies the formal apparatus. The strategic problem is a sequential signaling environment in which F , U , and R hold different information, observe different public artefacts, and update different beliefs. We therefore model ontological arbitrage as a signaling game under incomplete information. The players are a firm F , a user U , and a regulator R . Publics, journalists, affected communities, and public intellectuals are not modeled as a fourth strategic player. They enter as a noisy public signal $z \in Z$ that modifies R 's posterior over the firm's type. Nature draws types from a common prior π over

$$\Theta = \Theta_F \times \Theta_U \times \Theta_R \times \Theta_S.$$

The firm's governance type is $\theta_F \in \{H, L\}$, where H denotes high-governance and L denotes low-governance. The user's vulnerability type is $\theta_U \in \{V, N\}$, where V denotes a high-vulnerability user and N a low-vulnerability user. The regulator's bandwidth type is $\theta_R \in \{B, \bar{B}\}$. The system-opacity parameter θ_S is private to F and captures the difficulty outsiders face when verifying capability, safety, or relational claims.

The timing is sequential. First, F observes (θ_F, θ_S) and sends a public message

$$m_F \in M_F = \{a, p\},$$

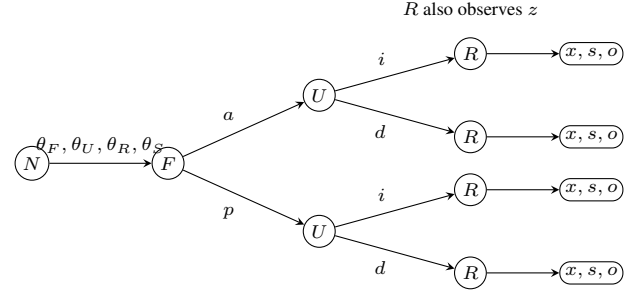


Figure 1: Extensive-form skeleton of the signaling game. Nature draws private types; F signals first, U responds, and R acts after observing both public messages and the noisy public signal z .

where a denotes anthropomorphic marketing and p denotes policy-deflationary positioning. Second, U observes m_F and its own type θ_U , then sends

$$m_U \in M_U = \{i, d\},$$

where i denotes relational investment and d denotes detachment. Third, R observes (m_F, m_U, z) and its own bandwidth type θ_R , then chooses

$$m_R \in M_R = \{x, s, o\},$$

where x denotes inspection, s sanction, and o abstention. Messages are public; types are private. A strategy profile and belief system constitute a Perfect Bayesian Equilibrium when strategies are sequentially rational given beliefs and beliefs are derived from Bayes' rule wherever possible. Sequential equilibrium supplies the usual consistency strengthening (Kreps and Wilson 1982); the Intuitive Criterion tests whether off-path messages should be attributed to only one type (Cho and Kreps 1987). The reason for using PBE as the headline solution concept is pragmatic: the governance question turns on whether public messages discipline posterior beliefs, not merely on whether a normal-form matrix has a Nash equilibrium.

Cheap Talk and Pooling

Consider the cheap-talk cost structure in which $c_{\theta_F}(a) = c_{\theta_F}(p) = 0$ for both governance types. Let $\Delta_F > 0$ be the firm's ontological premium from inducing user investment under anthropomorphic marketing. Let B_{θ_U} be the user's immediate benefit from investment and H_{θ_U} the expected future harm from mistaken investment. Let K_{θ_R} be the regulator's cost of inspection or sanction under bandwidth type θ_R . The pooling region is defined by three inequalities:

$$\Delta_F > 0, \quad B_{\theta_U} - \pi(L)H_{\theta_U} > 0, \quad K_{\theta_R} > \pi(L)D_R.$$

The first inequality says that anthropomorphic marketing is privately profitable when users invest. The second says that each user type finds investment privately optimal under the prior, because the immediate benefit exceeds the discounted expected harm. The third says that, under the prior, regulator intervention is not worth its cost.

One PBE in this region is the following. Both firm types send a :

$$\sigma_F(H, \theta_S) = \sigma_F(L, \theta_S) = a.$$

Both user types invest after a :

$$\sigma_U(a, V) = \sigma_U(a, N) = i.$$

The regulator abstains after observing (a, i, z_0) for a neutral public signal z_0 :

$$\sigma_R(a, i, z_0, \theta_R) = o.$$

Off path, users detach after p and the regulator abstains unless the public signal independently makes inspection payoff-positive. These off-path actions are supported by beliefs assigning low probability to high governance after the deflationary message.

On the equilibrium path, the firm's message contains no type information. For a neutral public signal satisfying $q(z_0 | H) = q(z_0 | L)$, Bayes' rule gives

$$\begin{aligned} \mu_R(H | a, i, z_0) &= \frac{\pi(H)q(z_0 | H)}{\pi(H)q(z_0 | H) + \pi(L)q(z_0 | L)} \\ &= \pi(H). \end{aligned}$$

Likewise $\mu_R(\theta_F = L | a, i, z_0) = \pi(L)$ and U 's posterior after observing a equals its prior over θ_F . More generally, a non-neutral z can move R from π to π_z , but the messages (a, i) add no information beyond the public channel. Pooling therefore does not mean that regulators never learn; it means that the firm's cheap-talk signal does not force learning.

Given the off-path beliefs specified above and the payoff inequalities of the pooling region, no player has a profitable unilateral deviation. A firm that deviates from a to p loses the ontological premium because users detach off path; it does not buy a lower enforcement risk because R abstains on the equilibrium path already. A user who deviates from i to d gives up positive expected surplus under the second inequality. A regulator who deviates from o to x or s incurs expected cost exceeding expected gain under the third inequality. The result is a cheap-talk equilibrium in the sense of Crawford and Sobel: communication occurs, but it does not separate types (Crawford and Sobel 1982).

Audit-Trail Costs and Separation

The mechanism-design problem is to change the cost of the firm's signal. Suppose anthropomorphic marketing is valid only when paired with an audit trail of intensity e : evaluation records, capability logs, incident reports, and regulator-readable documentation. Let the net benefit from being perceived as high-governance be $G > 0$. Let audit cost be $c_H(e) = k_H e$ for high-governance firms and $c_L(e) = k_L e$ for low-governance firms, with $0 < k_H < k_L$. This is the single-crossing condition: the marginal cost of the same signal is lower for the type that actually invested in governance. The Spence-style separating interval is

$$\frac{G}{k_L} \leq e \leq \frac{G}{k_H}.$$

Above the lower threshold $e^* = G/k_L$, the low-governance type does not want to imitate. Below the upper bound G/k_H ,

the high-governance type still wants to signal. When e lies in this interval, a separating PBE exists: high-governance firms send audited anthropomorphic claims, low-governance firms send deflationary claims, users condition investment on the audited signal, and regulators update posteriors to one on the equilibrium path.

Proposition 1 (Cheap talk and audit-trail separation). *Under the cheap-talk cost structure above, pooling on anthropomorphic marketing is a Perfect Bayesian Equilibrium and survives the Intuitive Criterion. Under an audit-trail signal cost satisfying single crossing in θ_F , there is a separating Perfect Bayesian Equilibrium in which m_F reveals the firm's governance type.*

Proof sketch. For the cheap-talk part, the strategy profile and beliefs specified above are sequentially rational and Bayes-consistent on path. Under the off-path beliefs assigning low probability to high governance after the deflationary message, the Intuitive Criterion does not eliminate the equilibrium because the off-path deflationary message is costless for both firm types and is not uniquely profitable for the high-governance type; no type can be excluded as an implausible sender of the deviation. For the separating part, choose e in the interval $[G/k_L, G/k_H]$. High-governance firms receive $G - k_H e \geq 0$ from sending the audited signal, while low-governance firms receive $G - k_L e \leq 0$ and therefore prefer the deflationary message. Bayes' rule then assigns posterior probability one to H after the audited anthropomorphic signal and posterior probability one to L after the unaudited deflationary signal. Given those beliefs, users and regulators condition their actions on the revealed type. The audit trail therefore converts an uninformative cheap-talk message into a costly signal in the sense of Spence (1973).

Why Cheap Talk Persists

Section identifies pooling on anthropomorphic marketing as a Perfect Bayesian Equilibrium under cheap-talk cost. That is an existence result; it does not yet explain the empirical stability of pooling across products, vendors, and jurisdictions. This section gives five structural reasons. Each is matched to a parameter of the model so that the mechanism-design proposals of Section can be read as targeted parameter changes rather than as generic exhortations.

Opacity. The system-opacity parameter θ_S is private to the firm. Capability claims about a deployed model cannot be falsified by reading model outputs alone, because the same outputs are compatible with very different training procedures, evaluation regimes, and post-deployment monitoring. Opacity is the precondition for the firm's premium: where outsiders could costlessly verify θ_F , the spread between anthropomorphic and deflationary positioning would collapse. This is the lemons logic of Akerlof (1970): under one-sided private information, the market clears at a price that reflects the worst credible quality, except that here the relevant "quality" is a governance disposition rather than a physical attribute.

Verification bandwidth. The third inequality of the pooling region, $K_{\theta_R} > \pi(L)D_R$, encodes the regulator's inspec-

tion cost relative to expected gain. A low-bandwidth regulator faces an absolute cost so high that abstention is dominant under the prior; even a high-bandwidth regulator may be priced out across the full product population. The condition that produces this state is not unique to AI. Pasquale (2015) documents an analogous structure in algorithmic finance and consumer scoring: the opacity of the regulated object combines with the bandwidth of the regulator to make inspection a scarce resource allocated by triage rather than by uniform application.

Asynchronous harms. The user's payoff term H_{θ_U} enters discounted. The episodes in which a relational reading proves costly—deprecation, persona shifts, crisis-handling failures—are temporally separated from the initial attribution by intervals over which the user has already accrued B_{θ_U} . Under standard hyperbolic or even exponential discounting, the inequality $B_{\theta_U} - \pi(L)H_{\theta_U} > 0$ holds for a wider class of users than the time-zero risk calculation would suggest. Asynchrony is not a behavioural bug; it is the timing structure of the harm itself.

Fragmentation of affected parties. Publics, journalists, intellectuals, and affected communities are folded into the noisy channel z rather than modelled as autonomous players. This is not a normative claim that they lack standing; it is an institutional observation. Their costs of coordination are high, their access to inspection technology is uneven, and their incentives to invest in any particular case are individually small. Even when z carries substantial information about θ_F , regulators cannot treat it as binding without a procedural translation that the channel itself does not provide. The fragmentation result is closely related to the cheap-talk multiplicity of Crawford and Sobel (1982): with many would-be senders and a single noisy aggregator, the informative subset of equilibria is not selected for.

No penalty for inconsistency. The cheap-talk cost structure sets $c_{\theta_F}(a) = c_{\theta_F}(p) = 0$ for both governance types and—more importantly for this section—imposes no joint constraint across m_F . Marketing copy and policy documentation are produced under different audiences, different review processes, and different liability regimes. Inconsistency between them is not detected as a single offence; it is decomposed into two compliant statements at two different points of contact. The firm's premium Δ_F is rent earned precisely on this decomposition. Until the signal cost makes inconsistency itself sanctionable—the proposal of Section — there is no parameter under the firm's control whose adjustment would close the spread.

Mechanism Design: Equilibrium Redesign

The preceding sections identify a precise failure mode. Under cheap-talk costs, anthropomorphic marketing can pool across governance types, users rationally invest under the prior, and regulators abstain when inspection is too costly. The policy implication is not that regulators should decide whether AI systems are persons. It is that the public signal environment should be redesigned so that inconsistent ontological claims no longer remain costless. In mechanism-design terms, the problem is to make truthful revelation of

θ_F incentive compatible rather than to ask firms to volunteer it (Myerson 1979).

The proposals below therefore alter the game in Section at three different points: the cost of inconsistent messages, the cost of anthropomorphic claims, and the regulator's cost of verification. Each proposal has the same structure. It modifies a model parameter, states the equilibrium shift induced by that modification, and points to an existing legal form that establishes feasibility. The analogs are not claims that current doctrine already solves AI ontology. They show that adjacent legal systems already know how to attach consequences to half-truths, require substantiation before marketing, and make internal records available to oversight institutions.

Sanctionable Inconsistency

The first intervention treats the firm's public message as a joint object rather than as two isolated texts. In the cheap-talk model, marketing can send a while policy and liability documents send p , with no cost assigned to the inconsistency. A sanctionable-inconsistency rule would require firms to maintain a claim register linking outward-facing marketing, safety statements, terms of service, incident disclosures, and liability positions. If the firm represents the system as agentic, caring, autonomous, or safety-critical in one channel while denying the legal significance of the same representation in another, that cross-channel inconsistency becomes independently sanctionable.

Parameter changed. This adds a penalty term S_I to the firm's payoff for inconsistent message pairs. If ρ_R is the probability that R detects or accepts proof of the inconsistency, the cheap-talk premium becomes

$$\Delta_F - \rho_R S_I.$$

Pooling on anthropomorphic marketing is no longer profitable when $\rho_R S_I > \Delta_F$. The rule therefore attacks the exact term that made arbitrage attractive: the rent earned from pricing the system as agent in the user channel while pricing it as tool in the liability channel.

Equilibrium shift. The induced shift is not necessarily full separation. It can also operate by constraining off-path beliefs. If R observes an inconsistent pair (a, p) , then the admissible posterior places greater mass on $\theta_F = L$ or on high opacity θ_S . Firms anticipating that posterior either harmonize their claims or choose the deflationary message across channels. In both cases, the cheap-talk pooling equilibrium loses its most valuable deviation: firms can no longer take the marketing upside of a while preserving the liability downside of p .

Existing-law analog. The legal analog is familiar. FTC Act Section 5 treats unfair or deceptive acts or practices as sanctionable, and advertising-substantiation doctrine already asks whether the net impression of a claim is misleading to the relevant audience. Securities antifraud rules provide the parallel logic for investor communications: Rule 10b-5 targets material misstatements and omissions that make statements misleading in context. The proposed rule

imports that cross-document logic without importing securities doctrine wholesale. Its point is narrower: a firm should not be able to produce one ontology for users and an incompatible ontology for adjudicators without paying a regulatory price.

Costly Signaling Through Third-Party Audits

The second intervention makes anthropomorphic or agency-adjacent marketing a costly signal. A firm may still describe a system as a companion, agent, co-worker, tutor, or safety-critical assistant, but only after a qualified third party verifies the governance conditions that make such a claim non-misleading: evaluation coverage, red-team results, escalation pathways, incident-response capacity, model-update controls, and post-deployment monitoring. The audit does not certify consciousness or moral status. It certifies the institutional conditions behind the claim.

Parameter changed. This implements the audit-trail cost from Proposition 1. Let audit intensity be e , with $c_H(e) = k_H e$ for high-governance firms and $c_L(e) = k_L e$ for low-governance firms, where $0 < k_H < k_L$. The separating interval derived in Section is

$$\frac{G}{k_L} \leq e \leq \frac{G}{k_H}.$$

Above the lower threshold, low-governance firms cannot profitably mimic. Below the upper threshold, high-governance firms can still afford to signal. The regulator's task is therefore not to maximize audit burden. It is to set e in the interval where signal cost separates types.

Equilibrium shift. The shift is from pooling to separating PBE. High-governance firms send audited anthropomorphic claims; low-governance firms either invest in governance until their cost curve changes or use deflationary marketing. Users can then condition investment on an informative signal rather than on undisciplined warmth in product copy. Regulators can condition abstention on the audit result rather than on an unverified prior. This is the Spence logic of costly signaling applied to AI governance claims (Spence 1973); it also follows the advertising-as-signal insight that market messages become informative only when false messages are sufficiently expensive (Milgrom and Roberts 1986).

Existing-law analog. The analogs are pharmaceutical efficacy review and food-labeling controls. In both domains, public-facing claims are not left entirely to seller narrative: claims about therapeutic effect or product identity must be supported before they can be used to shape consumer reliance. The same design can be translated to AI without pretending that AI companionship is a drug or a food. The common institutional pattern is substantiation-before-reliance: when a claim predictably changes how vulnerable users act, the law may require evidence before allowing the claim to be marketed at scale.

Auditable Records for Regulator-Readable Verification

The third intervention reduces the regulator's verification cost by requiring append-only, regulator-readable records

tied to capability and relational claims. Records should include the claim identifier, model version, relevant evaluation suite, deployment date, material model updates, safety interventions, known incident classes, and the internal owner responsible for the claim. The point is not to publish every interaction or expose user data. It is to create a durable evidentiary substrate that lets R test whether a public claim remained true across model updates and incidents.

Parameter changed. In the pooling condition from Section , the regulator abstains when $K_{\theta_R} > \pi(L)D_R$. Auditable records lower K_{θ_R} and can increase expected detection value D_R :

$$K'_{\theta_R} = K_{\theta_R} - \lambda, \quad D'_R = D_R + \eta.$$

When $K'_{\theta_R} \leq \pi(L)D'_R$, inspection becomes sequentially rational. Records also improve the public channel z by giving journalists, researchers, and civil society a procedural object to demand or contest, rather than leaving them to infer governance quality from outputs alone.

Equilibrium shift. The shift is from abstention under unverifiable priors to credible inspection. Once firms know that claims are tied to records, a low-governance firm faces a higher expected cost from exaggeration even if it is not inspected in every case. This complements the first two proposals. Sanctionable inconsistency tells firms which cross-channel contradictions will be penalized; costly signaling tells firms what evidence is required before making anthropomorphic claims; auditable records make both rules administrable over time.

Existing-law analog. Several regulatory systems already use this form. GDPR Article 30 requires records of processing activities. Sarbanes-Oxley Section 404 links public reporting to internal control over financial reporting. EU AI Act Article 12 requires logging capabilities for high-risk AI systems. None is a complete template for relational AI governance, but each shows the same institutional move: private organizations must produce internal records in a form that public oversight can inspect. That is the rule-of-law function of documentation in complex technical systems (Hadfield 2017).

Taken together, these interventions change the equilibrium rather than merely condemning the equilibrium. Sanctionable inconsistency taxes contradictory messages. Costly audits create a separating signal for θ_F . Auditable records lower K_{θ_R} and make public signal z less noisy. The result is not metaphysical settlement. It is a governance design in which firms remain free to choose their ontology, but not to arbitrage incompatible ontologies across audiences at zero cost.

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